

RAYAN KHAN

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EDUCATION

Rutgers University

Bachelor of Science, Aerospace and Mechanical Engineering

GPA: 3.54 | Honors: Dean's List | Relevant Coursework: Thermodynamics, Fluid Mechanics, Dynamic Systems and Controls

New Brunswick, NJ

Expected May 2028

EXPERIENCE

L'Oréal USA

SPARK Program – Mechanical Engineering Intern

Clark, NJ

June 2026 – August 2026

- Applied SolidWorks structural analysis to redesign load-bearing assemblies for 3+ product lines, reducing material mass by ~12%
- Executed 50+ compression, filling, and drop durability tests; logged and analyzed failure mode data in MATLAB
- Coordinated with 8+ engineers in Supply Chain and Operations on subsystem design reviews, achieving ~10% reduction in material cost

Rutgers University — Low-Reynolds-Number Aerodynamics Lab

Undergraduate Researcher

New Brunswick, NJ

October 2025 – Present

- Developed an automated OpenFOAM/Python CFD pipeline computing CL, CD, CM for NACA 0012/2412/4412 at $Re = 2 \times 10^5$
- Conducted 87+ steady RANS simulations from -4° to 20° AoA with 0.5° stall resolution; validated against XFOIL within 4% error
- Engineered a structured C-mesh blockMesh topology resolving pseudo-3D artifacts; validated pressure fields via ParaView

Rocket Launch Safety Simulator

Designer & Programmer

New Brunswick, NJ

2025 – 2026

- Engineered an Arduino-based closed-loop launch abort system integrating ultrasonic proximity scanning and temperature sensing to autonomously abort unsafe countdowns
- Implemented non-blocking sensor fusion enabling real-time abort detection within <1 second; mitigated servo-induced noise
- Architected a 5-state FSM (OFF → ARMED → COUNTDOWN → SUCCESS/ABORT) with synchronized LED, buzzer, and LCD outputs

PROJECTS

Two-Body Orbital Propagator with J2 Perturbations

Programmer

May 2026- June 2026

- Built a Python orbital mechanics engine implementing RK45 integration of the two-body equations; validated J2 nodal regression to 0.4%
- Propagated ISS, GEO, and Molniya orbits from real parameters; generated 3D trajectory plots validating correct orbital behavior
- Implemented a Hohmann transfer ΔV calculator and GEO altitude; all orbital element round-trip conversions pass to $<1 \times 10^{-8}$ relative error

FEA Structural Analysis Tool

Programmer

March 2026- May 2026

- Built a linear static finite element solver from scratch in Python for 3D structural stress without any commercial FEA package
- Implemented Tet4 and quadratic Tet10 elements with Gauss quadrature; verified via the rigid-body eigenvalue spectrum
- Assembled sparse global stiffness with direct and conjugate-gradient solvers; applied Dirichlet constraints and consistent nodal loads

LEADERSHIP & ACTIVITIES

Accessible 4 All

Director of Fundraising / Lead Engineer

Summer 2026

- Fundraised over \$3,500 for the nonprofit organization through a series of donor outreach efforts and hands-on internal affairs work
- Co-founded a registered 501(c)(3) engineering nonprofit designing & modifying adaptive devices for children with disabilities
- Led multiple teams ranging from 5–10 to 20+ people to modify toys for children with disabilities, prioritizing strategic thinking

AIAA Design/Build/Fly (DBF) Team

Aerostructures Engineer

Summer 2026

- Engineered and designed front and rear wing assemblies in SolidWorks to competition regulations, achieving 120 N downforce at 60 km/h
- Engineered and designed front and rear wing assemblies in SolidWorks to FSAE regulations, achieving 120 N downforce at 60 km/h
- Co-founded a registered 501(c)(3) engineering nonprofit designing & modifying adaptive devices for children with disabilities
- Lead multiple teams of people ranging from 5-10 to 20+ to modify toys for children with disabilities prioritizing strategic thinking

SKILLS

Software & CAD: SolidWorks, Fusion 360, ANSYS Fluent, MATLAB, Python, OpenFOAM, ParaView, Git/GitHub

CFD & Simulation: k- ω SST RANS, blockMesh C-mesh, Mesh Quality Analysis, Aerodynamic Coefficient Extraction, snappyHexMesh

Engineering & Analysis: Computational Fluid Dynamics, FEA, Structural Analysis, Failure Mode Analysis Signal Processing

Languages: English (Native), Urdu (Proficient), Hindi (Limited Working Proficiency)